

EXTENSION 1

Stage 6

Proof (Ext1)

P1 Induction (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 78 minutes (based on allocation of 1.5 minutes per mark)



P1 Proof

1%

2%

3%

4%

5%

Sum of Series

Divisibility

*SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.

Questions

1. Proof, EXT1 P1 2014 HSC 13a

Use mathematical induction to prove that $2^n + (-1)^{n+1}$ is divisible by 3 for all integers $n \geq 1$. (3 marks)

2. Proof, EXT1 P1 2015 HSC 13c

Prove by mathematical induction that for all integers $n \geq 1$,
$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}.$$
 (3 marks)

3. Proof, EXT1 P1 2018 HSC 13a

Prove by mathematical induction that, for $n \geq 1$,
$$2 - 6 + 18 - 54 + \dots + 2(-3)^{n-1} = \frac{1 - (-3)^n}{2}.$$
 (3 marks)

4. Proof, EXT1 P1 2020 HSC 12a

Use the principle of mathematical induction to show that for all integers $n \geq 1$,
$$1 \times 2 + 2 \times 5 + 3 \times 8 + \dots + n(3n - 1) = n^2(n + 1).$$
 (3 marks)

5. Proof, EXT1 P1 2023 HSC 12b

Use mathematical induction to prove that
$$(1 \times 2) + (2 \times 2^2) + (3 \times 2^3) + \dots + (n \times 2^n) = 2 + (n - 1)2^{n+1}$$
for all integers $n \geq 1$. (3 marks)

6. Proof, EXT1 P1 2024 HSC 12d

Use mathematical induction to prove that $2^{3n} + 13$ is divisible by 7 for all integers $n \geq 1$. (3 marks)

7. Proof, EXT1 P1 2019 HSC 14a

Prove by mathematical induction that, for all integers $n \geq 1$,
$$1(1!) + 2(2!) + 3(3!) + \dots + n(n!) = (n + 1)! - 1.$$
 (3 marks)

8. Proof, EXT1 P1 2016 HSC 14a

i. Show that $4n^3 + 18n^2 + 23n + 9$ can be written as

$(n + 1)(4n^2 + 14n + 9)$. (1 marks)

ii. Using the result in part (i), or otherwise, prove by mathematical induction that, for $n \geq 1$,

$1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2n - 1)(2n + 1) = \frac{1}{3}n(4n^2 + 6n - 1)$. (3 marks)

9. Proof, EXT1 P1 2007 HSC 4b

Use mathematical induction to prove that $7^{2n-1} + 5$ is divisible by 12, for all integers $n \geq 1$. (3 marks)

10. Proof, EXT1 P1 2008 HSC 3b

Use mathematical induction to prove that, for integers $n \geq 1$,

$1 \times 3 + 2 \times 4 + 3 \times 5 + \dots + n(n + 2) = \frac{n}{6}(n + 1)(2n + 7)$. (3 marks)

11. Proof, EXT1 P1 2017 HSC 14a

Prove by mathematical induction that $8^{2n+1} + 6^{2n-1}$ is divisible by 7, for any integer $n \geq 1$. (3 marks)

12. Proof, EXT1 P1 2021 HSC 12c

Use mathematical induction to prove that

$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \dots + \frac{1}{n(n + 1)(n + 2)} = \frac{1}{4} - \frac{1}{2(n + 1)(n + 2)}$

for all integers $n \geq 1$. (3 marks)

13. Proof, EXT1 P1 2022 HSC 12f

Use mathematical induction to prove that $15^n + 6^{2n+1}$ is divisible by 7 for all integers $n \geq 0$. (3 marks)

14. Proof, EXT1 P1 2012 HSC 12a

Use mathematical induction to prove that $2^{3n} - 3^n$ is divisible by 5 for $n \geq 1$. (3 marks)

15. Proof, EXT1 P1 2011 HSC 6a

Use mathematical induction to prove that for $n \geq 1$,

$1 \times 5 + 2 \times 6 + 3 \times 7 + \dots + n(n + 4) = \frac{1}{6}n(n + 1)(2n + 13)$. (3 marks)

16. Proof, EXT1 P1 SM-Bank 3

Prove by mathematical induction that $(3n + 1)7^n - 1$ is divisible by 9 for integral $n \geq 1$. (3 marks)

17. Proof, EXT1 P1 2010 HSC 7a

Prove by induction that

$47^n + 53 \times 147^{n-1}$

is divisible by 100 for all integers $n \geq 1$. (3 marks)

Worked Solutions

1. Proof, EXT1 P1 2014 HSC 13a

Prove $2^n + (-1)^{n+1}$

is divisible by 3 for integers $n \geq 1$

If $n = 1$

$$\begin{aligned} 2^1 + (-1)^2 &= 2 + 1 \\ &= 3 \quad (\text{which is divisible by 3}) \\ \therefore \text{ True for } n &= 1 \end{aligned}$$

Assume true for $n = k$

$$\begin{aligned} \text{i.e. } 2^k + (-1)^{k+1} &= 3P \quad (P \text{ integer}) \\ 2^k &= 3P - (-1)^{k+1} \quad \dots (*) \end{aligned}$$

Prove true for $n = k + 1$

$$\begin{aligned} 2^{k+1} + (-1)^{k+1+1} &= 2 \cdot 2^k + (-1)^{k+2} \\ &= 2 \left(3P - (-1)^{k+1} \right) + (-1)^{k+2} \quad \dots \text{from } (*) \\ &= 6P - 2(-1)^{k+1} - 1(-1)^{k+1} \\ &= 6P - 3(-1)^{k+1} \\ &= 3 \left(2P - (-1)^{k+1} \right) \end{aligned}$$

$\dots$ which is divisible by 3

$\Rightarrow$ True for $n = k + 1$

$\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

2. Proof, EXT1 P1 2015 HSC 13c

Prove for $n \geq 1$

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$$

If $n = 1$

$$\text{LHS} = \frac{1}{2!} = \frac{1}{2}$$

$$\text{RHS} = 1 - \frac{1}{2!} = 1 - \frac{1}{2} = \frac{1}{2}$$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$$\text{i.e. } \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$$

Prove true for $n = k + 1$

$$\text{i.e. } \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = 1 - \frac{1}{(k+2)!}$$

$$\begin{aligned} \text{LHS} &= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \\ &= 1 - \left(\frac{(k+2) - (k+1)}{(k+1)!(k+2)} \right) \\ &= 1 - \frac{1}{(k+2)!} \quad \dots \text{as required} \end{aligned}$$

$\Rightarrow$ True for $n = k + 1$

$\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

3. Proof, EXT1 P1 2018 HSC 13a

If $n = 1$,

$$\text{LHS} = 2(-3)^0 = 2$$

$$\text{RHS} = \frac{1 - (-3)^1}{2} = 2$$

∴ True for $n = 1$

Assume true for $n = k$

$$2 - 6 + 18 - 54 + \dots + 2(-3)^{k-1} = \left(1 - (-3)^k\right)$$

Prove true for $n = k + 1$

$$\text{i.e. } 2 - 6 + \dots + 2(-3)^{k-1} + 2(-3)^k = \frac{1 - (-3)^{k+1}}{2}$$

$$\begin{aligned} \text{LHS} &= \underbrace{2 - 6 + \dots + 2(-3)^{k-1}}_{\text{Sum of GP, } r = -3} + 2(-3)^k \\ &= \frac{2\left(1 - (-3)^k\right)}{1 + 3} + 2(-3)^k \\ &= \frac{1 - (-3)^k + 4(-3)^k}{2} \\ &= \frac{1 + 3(-3)^k}{2} \\ &= \frac{1 - (-3)(-3)^k}{2} \\ &= \frac{1 - (-3)^{k+1}}{2} \\ &= \text{RHS} \end{aligned}$$

⇒ True for $n = k + 1$

∴ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

4. Proof, EXT1 P1 2020 HSC 12a

Prove

$$1 \times 2 + 2 \times 5 + 3 \times 8 + \dots + n(3n - 1) = n^2(n + 1)$$

Prove true for $n = 1$:

$$\text{LHS} = 1(3 \times 1 - 1) = 2$$

$$\text{RHS} = 1^2(1 + 1) = 2$$

Assume true for $n = k$:

$$\text{i.e. } 1 \times 2 + 2 \times 5 + 3 \times 8 + \dots + k(3k - 1) = k^2(k + 1)$$

Prove true for $n = k + 1$:

$$\text{i.e. } 1 \times 2 + 2 \times 5 + \dots + k(3k - 1) + (k + 1)(3k + 2) = (k + 1)^2(k + 2)$$

$$\begin{aligned} \text{LHS} &= 1 \times 2 + \dots + k(3k - 1) + (k + 1)(3k + 2) \\ &= k^2(k + 1) + (k + 1)(3k + 2) \\ &= (k + 1)(k^2 + 3k + 2) \\ &= (k + 1)(k + 1)(k + 2) \\ &= (k + 1)^2(k + 2) \\ &= \text{RHS} \end{aligned}$$

⇒ True for $n = k + 1$

∴ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

5. Proof, EXT1 P1 2023 HSC 12b

Prove true for $n = 1$:
 $LHS = 1 \times 2 = 2$
 $RHS = 2 + (1 - 1)2^2 = 2 = LHS$

Assume true for $n = k$:
 $(1 \times 2) + (2 \times 2^2) + \dots + (k \times 2^k) = 2 + (k - 1)2^{k+1}$

Prove true for $n = k + 1$:
 $(1 \times 2) + (2 \times 2^2) + \dots + (k \times 2^k) + (k + 1)2^{k+1} = 2 + k \times 2^{k+2}$

$LHS = 2 + (k - 1)2^{k+1} + (k + 1)2^{k+1}$
 $= 2 + 2^{k+1}(k - 1 + k + 1)$
 $= 2 + 2^{k+1}(2k)$
 $= 2 + k \times 2^{k+2}$
 $= RHS$

$\Rightarrow$ True for $n = k + 1$
 $\therefore$ Since true for $n = 1$, by PMI, true for integers $n \geq 1$.

6. Proof, EXT1 P1 2024 HSC 12d

Prove $2^{3n} + 13$ is divisible by 7 for $n \geq 1$.
If $n = 1$:
 $2^{3 \times 1} + 13 = 21 = 3 \times 7$
 $\therefore$ True for $n = 1$.

Assume true for $n = k$:
 $2^{3k} + 13 = 7P$ (where P is an integer)
 $\Rightarrow 2^{3k} = 7P - 13 \dots (1)$

Prove true for $n = k + 1$:

$2^{3(k+1)} + 13 = 2^{3k} \times 2^3 + 13$
 $= 8(2^{3k}) + 13$
 $= 8(7P - 13) + 13$ (see (1) above)
 $= 56P - 8 \times 13 + 13$
 $= 7(8P - 13)$

$\Rightarrow$ True for $n = k + 1$
 $\therefore$ Since true for $n = 1$, by PMI, true for integers $n \geq 1$.

7. Proof, EXT1 P1 2019 HSC 14a

Prove true for $n = 1$:
 $LHS = 1(1!) = 1$
 $RHS = (1 + 1)! - 1 = 2! - 1 = 1 = LHS$
 $\therefore$ True for $n = 1$

Assume true for $n = k$:
 $1(1!) + 2(2!) + \dots + k(k!) = (k + 1)! - 1$

Prove true for $n = k + 1$
i.e. $\underbrace{1(1!) + 2(2!) + \dots + k(k!)}_{(k+1)!-1} + (k + 1)(k + 1)! = (k + 2)! - 1$

$LHS = (k + 1)! - 1 + (k + 1)(k + 1)!$
 $= (k + 1)![1 + (k + 1)] - 1$
 $= (k + 1)!(k + 2) - 1$
 $= (k + 2)! - 1$

$\Rightarrow$ True for $n = k + 1$
 $\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

8. Proof, EXT1 P1 2016 HSC 14a

i.
$$\begin{aligned} \text{RHS} &= (n + 1)(4n^2 + 14n + 9) \\ &= 4n^3 + 14n^2 + 9n + 4n^2 + 14n + 9 \\ &= 4n^3 + 18n^2 + 23n + 9 \end{aligned}$$

ii. Prove $1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2n-1)(2n + 1)$

$$= \frac{1}{3}n(4n^2 + 6n-1)$$

If $n = 1$,
LHS = $(1)(3) = 3$
RHS = $\frac{1}{3}(1)(4 + 6-1) = 3$
 $\therefore$ True for $n = 1$

Assume true for $n = k$
i.e. $1 \times 3 + 3 \times 5 + \dots + (2k-1)(2k + 1)$

$$= \frac{1}{3}k(4k^2 + 6k-1)$$

Prove true for $n = k + 1$
i.e. $1 \times 3 + \dots + (2k + 1)(2k + 3)$

$$\begin{aligned} &= \frac{1}{3}(k + 1)\left[4(k + 1)^2 + 6(k + 1)-1\right] \\ &= \frac{1}{3}(k + 1)(4k^2 + 14k + 9) \\ &= \frac{1}{3}(4k^3 + 18k^2 + 23k + 9) \end{aligned}$$

$$\begin{aligned} \text{LHS} &= 1 \times 3 + \dots + (2k-1)(2k + 1) + (2k + 1)(2k + 3) \\ &= \frac{1}{3}k(4k^2 + 6k-1) + (2k + 1)(2k + 3) \\ &= \frac{1}{3}(4k^3 + 6k^2-k) + (4k^2 + 8k + 3) \\ &= \frac{1}{3}(4k^3 + 6k^2-k + 12k^2 + 24k + 9) \\ &= \frac{1}{3}(4k^3 + 18k^2 + 23k + 9) \\ &= \text{RHS} \dots \end{aligned}$$

$\Rightarrow$ True for $n = k + 1$
 $\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

9. Proof, EXT1 P1 2007 HSC 4b

Prove $7^{2n-1} + 5$ is divisible by 12, for $n \geq 1$
If $n = 1$
 $7^{2-1} + 5 = 7 + 5 = 12$ (divisible by 12)
 $\therefore$ True for $n = 1$

Assume true for $n = k$
i.e. $7^{2k-1} + 5 = 12\text{N}$ (N is an integer)

$$7^{2k-1} = 12\text{N}-5 \dots (1)$$

Prove true for $n = k + 1$

$$\begin{aligned} 7^{2(k+1)-1} + 5 &= 7^{2k+1} + 5 \\ &= 7^2 \cdot 7^{2k-1} + 5 \\ &= 49(12\text{N} - 5) + 5 \quad (\text{from (1) above}) \\ &= 49 \cdot 12\text{N} - 245 + 5 \\ &= 49 \cdot 12\text{N} - 240 \\ &= 12(49\text{N} - 20) \end{aligned}$$

$\dots$ which is divisible by 12

$\Rightarrow$ True for $n = k + 1$
 $\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

10. Proof, EXT1 P1 2008 HSC 3b

Prove $1 \times 3 + 2 \times 4 + 3 \times 5 + \dots + n(n + 2)$
 $= \frac{n}{6}(n + 1)(2n + 7)$ for $n \geq 1$

If $n = 1$
 $\text{LHS} = 1 \times 3 = 3$
 $\text{RHS} = \frac{1}{6}(2)(9) = 3 = \text{LHS}$

$\therefore$ True for $n = 1$

Assume true for $n = k$
i.e. $1 \times 3 + 2 \times 4 + \dots + k(k + 2)$
 $= \frac{k}{6}(k + 1)(2k + 7)$

Prove true for $n = k + 1$
i.e. $1 \times 3 + 2 \times 4 + \dots + k(k + 2) + (k + 1)(k + 3)$
 $= \frac{(k + 1)}{6}(k + 2)(2k + 9)$

$\text{LHS} = \frac{k}{6}(k + 1)(2k + 7) + (k + 1)(k + 3)$
 $= \frac{1}{6}(k + 1)[k(2k + 7) + 6(k + 3)]$
 $= \frac{1}{6}(k + 1)[2k^2 + 7k + 6k + 18]$
 $= \frac{1}{6}(k + 1)[2k^2 + 13k + 18]$
 $= \frac{1}{6}(k + 1)(k + 2)(2k + 9) \quad \dots \text{as required}$

$\Rightarrow$ True for $n = k + 1$
 $\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

11. Proof, EXT1 P1 2017 HSC 14a

Prove $8^{2n+1} + 6^{2n-1}$ is divisible
by 7 for integers $n \geq 1$
If $n = 1$,
 $8^3 + 6^1 = 518 = 74 \times 7$
 $\therefore$ True for $n = 1$

Assume true for $n = k$
i.e. $8^{2k+1} + 6^{2k-1} = 7P$ (where P is an integer)
 $8^{2k+1} = 7P - 6^{2k-1}$

Prove true for $n = k + 1$
 $8^{2k+3} + 6^{2k-1} = 64 \cdot 8^{2k+1} + 36 \cdot 6^{2k-1}$
 $= 64(7P - 6^{2k-1}) + 36 \cdot 6^{2k-1}$
 $= 64 \cdot 7P - 64 \cdot 6^{2k-1} + 36 \cdot 6^{2k-1}$
 $= 64 \cdot 7P - 28 \cdot 6^{2k-1}$
 $= 7(64P - 4 \cdot 6^{2k-1})$

$\dots$ which is divisible by 7.

$\Rightarrow$ True for $n = k + 1$
 $\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

12. Proof, EXT1 P1 2021 HSC 12c

Prove for $n \geq 1$,

$$\frac{1}{1 \times 2 \times 3} + \dots + \frac{1}{n(n+1)(n+2)} = \frac{1}{4} - \frac{1}{2(n+1)(n+2)}$$

If $n = 1$:

$$\text{LHS} = \frac{1}{1 \times 2 \times 3} = \frac{1}{6}$$

$$\text{RHS} = \frac{1}{4} - \frac{1}{2 \times 2 \times 3} = \frac{1}{4} - \frac{1}{12} = \frac{1}{6}$$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$$\text{i.e. } \frac{1}{1 \times 2 \times 3} + \dots + \frac{1}{k(k+1)(k+2)} = \frac{1}{4} - \frac{1}{2(k+1)(k+2)}$$

Prove true for $n = k + 1$:

$$\text{i.e. } \frac{1}{1 \times 2 \times 3} + \dots + \frac{1}{(k+1)(k+2)(k+3)} = \frac{1}{4} - \frac{1}{2(k+2)(k+3)}$$

$$\begin{aligned} \text{LHS} &= \frac{1}{1 \times 2 \times 3} + \dots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \\ &= \frac{1}{4} - \frac{1}{2(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \\ &= \frac{1}{4} - \left(\frac{k+3-2}{2(k+1)(k+2)(k+3)} \right) \\ &= \frac{1}{4} - \frac{k+1}{2(k+1)(k+2)(k+3)} \\ &= \frac{1}{4} - \frac{1}{2(k+2)(k+3)} \\ &= \text{RHS} \end{aligned}$$

$\Rightarrow$ True for $n = k + 1$

$\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

13. Proof, EXT1 P1 2022 HSC 12f

Prove $15^n + 6^{2n+1}$ is divisible by 7 for $n \geq 0$

If $n = 0$:

$$15^0 + 6^1 = 7$$

$\therefore$ True for $n = 0$

Assume true for $n = k$

i.e. $15^k + 6^{2k+1} = 7P$ (where P is an integer)

$$\Rightarrow 6^{2k+1} = 7P - 15^k \dots (1)$$

Prove true for $n = k + 1$

$$\begin{aligned} 15^{k+1} + 6^{2k+3} &= 15 \cdot 15^k + 6^2 \cdot 6^{2k+1} \\ &= 15 \cdot 15^k + 36(7P - 15^k) \quad (\text{see (1)}) \\ &= 15 \cdot 15^k + 36 \cdot 7P - 36 \cdot 15^k \\ &= 36 \cdot 7P - 21 \cdot 15^k \\ &= 7(36P - 3 \cdot 15^k) \quad (\text{which is divisible by 7}) \end{aligned}$$

$\Rightarrow$ True for $n = k + 1$

$\therefore$ Since true for $n = 0$, by PMI, true for integers $n \geq 0$

14. Proof, EXT1 P1 2012 HSC 12a

Prove by induction

$2^{3n} - 3^n$ is divisible by 5 for $n \geq 1$

If $n = 1$

$$\begin{aligned} 2^{3n} - 3^n &= 2^3 - 3^1 \\ &= 5 \quad (\text{divisible by } 5) \\ \therefore \text{ True for } n &= 1 \end{aligned}$$

Assume true for $n = k$

$$\begin{aligned} \text{i.e. } 2^{3k} - 3^k &= 5P \quad (P \text{ integer}) \\ 3^k &= 2^{3k} - 5P \quad (1) \end{aligned}$$

Prove true for $n = k + 1$

$$\begin{aligned} 2^{3(k+1)} - 3^{k+1} &= 2^{3k+3} - 3 \cdot 3^k \\ &= 2^3 \cdot 2^{3k} - 3(2^{3k} - 5P) \quad (\text{from (1) above}) \\ &= 8 \cdot 2^{3k} - 3 \cdot 2^{3k} + 15P \\ &= 5 \cdot 2^{3k} + 15P \\ &= 5(2^{3k} + 3P) \end{aligned}$$

$\Rightarrow$ True for $n = k + 1$
 $\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$

IMPORTANT: Making 3^k or 2^{3k} the subject helps for the substitution required in the proof for $n + 1$. Important to state that P is an integer.

15. Proof, EXT1 P1 2011 HSC 6a

$$\begin{aligned} \text{Prove } 1 \times 5 + 2 \times 6 + 3 \times 7 + \dots + n(n+4) \\ = \frac{1}{6}n(n+1)(2n+13) \text{ for } n \geq 1 \end{aligned}$$

If $n = 1$

$$\text{LHS} = 1 \times 5 = 5$$

$$\text{RHS} = \frac{1}{6}(1)(2)(15) = \frac{30}{6} = 5 = \text{LHS}$$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$$\begin{aligned} \text{i.e. } 1 \times 5 + 2 \times 6 + 3 \times 7 + \dots + k(k+4) \\ = \frac{1}{6}k(k+1)(2k+13) \end{aligned}$$

Prove true for $n = k + 1$

$$\begin{aligned} \text{i.e. } 1 \times 5 + 2 \times 6 + \dots + k(k+4) + (k+1)(k+5) \\ = \frac{1}{6}(k+1)(k+2)(2k+15) \end{aligned}$$

$$\begin{aligned} \text{LHS} &= \frac{1}{6}k(k+1)(2k+13) + (k+1)(k+5) \\ &= \frac{1}{6}(k+1)[k(2k+13) + 6(k+5)] \\ &= \frac{1}{6}(k+1)[2k^2 + 13k + 6k + 30] \\ &= \frac{1}{6}k(k+1)(2k^2 + 19k + 30) \\ &= \frac{1}{6}k(k+1)(k+2)(2k+15) \\ &= \text{RHS} \dots \text{ as required} \end{aligned}$$

$\Rightarrow$ True for $n = k + 1$

$\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$.

MARKER'S COMMENT: Write out the statement to be proven. Transcription errors, poor setting out and inefficient algebraic approaches (factorise whenever possible) were common errors.

16. Proof, EXT1 P1 SM-Bank 3

Prove $(3n + 1)7^n - 1$ is divisible
by 9 for integers $n \geq 1$.
If $n = 1$
 $(3n + 1)7^n - 1 = (3 \times 1 + 1)7^1 - 1$
 $= 27$
 $\Rightarrow$ Divisible by 9 ($27/9 = 3$)

Assume true for $n = k$
i.e. $(3k + 1)7^k - 1 = 9P$ (P integer)
 $f(k) = (3k + 1)7^k - 1$
 $f(k + 1) = (3(k + 1) + 1)7^{k+1} - 1$
 $= (3k + 4)7^{k+1} - 1$

$f(k + 1) - f(k)$
 $= (3k + 4)7^{k+1} - 1 - [(3k + 1)7^k - 1]$
 $= 3k \cdot 7^{k+1} + 4 \cdot 7^{k+1} - 1 - [3k \cdot 7^k + 7^k - 1]$
 $= 21k \cdot 7^k + 28 \cdot 7^k - 1 - 3k \cdot 7^k - 7^k + 1$
 $= 18k \cdot 7^k + 27 \cdot 7^k$
 $= 9(2k \cdot 7^k + 3 \cdot 7^k)$ (divisible)

Since $f(k + 1) - f(k)$ is divisible by 9 and
 $f(k)$ is divisible by 9,

$\Rightarrow f(k + 1)$ is divisible by 9
Since true for $n = 1$, true for integral $n \geq 1$

IMPORTANT: In cases where substitution doesn't work, remember that $f(k + 1) - f(k)$ can be used to illustrate divisibility as well (Tony Lantry special).

17. Proof, EXT1 P1 2010 HSC 7a

Prove $47^n + 53 \times 147^{n-1}$ is divisible
by 100 for $n \geq 1$.
If $n = 1$
 $47^1 + 53 \times 147^{1-1} = 47 + 53$
 $= 100$ (divisible by 100)
 $\therefore$ True for $n = 1$

Assume true for $n = k$
i.e. $47^k + 53 \times 147^{k-1} = 100P$ (P integer)
 $47^k = 100P - (53 \times 147^{k-1}) \dots (1)$

Prove true for $n = k + 1$
 $47^{k+1} + 53 \times 147^k$
 $= 47 \times 47^k + 53 \times 147^k$
(substitute from (1) above)
 $= 47[100P - (53 \times 147^{k-1})] + 53 \times 147 \times 147^{k-1}$
 $= 4700P - 2491 \times 147^{k-1} + 7791 \times 147^{k-1}$
 $= 4700P + 5300 \times 147^{k-1}$
 $= 100(47P + 53 \times 147^{k-1})$

$\Rightarrow$ True for $n = k + 1$
 $\therefore$ Since true for $n = 1$, by PMI, true for integral $n \geq 1$

♦ Mean mark 51%.
IMPORTANT: Making 47^k the subject helps the substitution in the proof for $n + 1$. Students must state that P is an integer. **NOTE:** using $147^k = 147 \times 147^{k-1}$ develops proof for $n = k + 1$.